

KYC/eKYC

#	Bài toán cụ thể	Function / journey	Data signal	AI/DS/ML	Output	Được gì	Rủi ro / mất gì
1	eKYC exception review scoring	Sau khi eKYC engine xử lý xong	OCR confidence, face match, liveness, document quality, device/IP risk	Rule + ML-lite classifier	pass/review/reject suggestion	Định vị đúng: không làm lại eKYC, chỉ xử lý case không chắc chắn	Cần synthetic signals hợp lý
2	Form-vs-document mismatch detection	User nhập form + upload giấy tờ	Họ tên, DOB, ID number, address, OCR result	Rule + NLP/LLM structured comparison	mismatch flags + reason	Cụ thể, dễ demo, hợp KYC exception	Nếu chỉ exact match thì quá đơn giản
3	OCR confidence triage	Sau OCR giấy tờ	OCR confidence theo field, missing field, text noise	DS/rule + ML-lite	accept OCR / request retake / manual review	Học uncertainty handling	Không có OCR thật thì phải mô phỏng
4	Low-quality document retake decision	Upload CCCD/passport	blur, glare, crop, brightness, OCR confidence	CV-lite/rule/ML	retake_required / continue	Rất thực tế trong onboarding	Làm CV thật có thể quá sức

5	Document field extraction validation	Sau OCR/LLM extraction	extracted fields, schema validity, ID format, DOB format	LLM structured output + validation	valid/invalid fields	Học structured output + guardrails	Không nên chỉ gọi LLM mà thiếu validator
6	Name normalization and fuzzy matching	So sánh tên form/OCR/account	Unicode, dấu tiếng Việt, order tên, typo, OCR noise	NLP/fuzzy matching	match score + mismatch reason	Rất cụ thể với dữ liệu VN	Có thể không đủ ML nếu chỉ rule
7	Address consistency checker	So sánh địa chỉ form/OCR/proof	province/district text, normalized address, edit distance	NLP normalization + rule	address mismatch / uncertain	Học data cleaning thực tế	Address VN phức tạp, dễ mất thời gian
8	DOB / age eligibility check	Onboarding	DOB, document issue date, user-declared age	Rule + anomaly	eligible / review / reject	Nhỏ, rõ, dễ dùng làm submodule	Quá đơn giản nếu đứng riêng
9	Expired / invalid document detection	Document validation	expiry date, issue date, document type	Rule + date validation	expired / valid / review	Cần thiết trong KYC flow	Không có AI/ML rõ, nên làm module phụ
10	Face match uncertainty routing	Selfie vs ID portrait	face_match_score, capture quality, confidence threshold	ML-lite/rule	auto-pass / retake / review	Học thresholding và false positive/false negative	Không nên claim làm face recognition thật

11	Liveness retry recommendation	Selfie/liveness step	liveness_score, failed attempts, device, lighting quality	Rule + ML-lite	retry / manual review / reject	Function nhỏ, thực tế	Liveness thật khó; nên dùng simulated scores
12	Repeated failed liveness anomaly	Onboarding retry flow	số lần fail, thời gian retry, device/IP reuse	Anomaly detection	suspicious retry pattern	Gần risk, nhưng vẫn thuộc KYC	Cần synthetic behavior data
13	Duplicate identity detection	New user onboarding	same ID number, phone, email, bank account, device	Entity resolution + anomaly	possible duplicate user	Rất thực tế cho ví/ngân hàng	Dễ chuyển sang fraud; cần giữ scope onboarding
14	Multi-account device reuse detection	Onboarding/device binding	device_id, IP, phone, ID, session fingerprint	Graph/anomaly	suspicious cluster	Có graph angle, đáng học	Graph có thể khó nếu chưa quen
15	Synthetic identity risk scoring	New application	partial match identity, weak document confidence, new phone/email/device	ML/rule hybrid	synthetic identity risk	Bài khó, có chiều sâu	Data khó, dễ overclaim

16	Watchlist / sanctions false-positive triage	AML/KYC screening	name match score, DOB, nationality, watchlist hit type	NLP matching + rule + LLM summary	true/possible/false hit	Có compliance angle mạnh	Nhạy cảm, cần dùng synthetic watchlist
17	PEP match explanation assistant	PEP screening review	matched name, DOB, role, country, confidence	LLM summary + evidence extraction	review note + reason	Hợp AI assistant có evidence	Không được bịa; cần source/evidence rõ
18	Adverse media screening summarizer	Enhanced due diligence	news snippets, entity name, date, source credibility	NLP/LLM classification + summary	risk summary	Gần real KYC/AML analyst workflow	Cần web/news data; dễ phức tạp
19	KYC risk banding from mixed signals	Sau khi gom tín hiệu KYC	OCR, face, liveness, device, watchlist, duplicate, form quality	ML classifier/rule score	low/medium/high KYC risk	Gom được nhiều signal thành decision support	Cần tránh làm quá rộng
20	KYC case routing	Manual review queue	risk band, issue type, document type, reviewer skill	ML/ranking/rule	route to normal/senior/compliance	Có ops workflow rõ	Cần giả lập reviewer/team

21	KYC reviewer checklist generator	Reviewer mở case	risk flags, mismatch type, missing fields	LLM structured output	checklist kiểm tra	Hợp AI agent/eval	Nếu không có schema/eval thì dễ thành chatbot
22	KYC audit summary generation	Sau review	input signals, reviewer decision, checklist result	LLM structured summary	audit note chuẩn hóa	Compliance mindset, dễ demo	LLM phải grounded vào facts
23	KYC rejection reason classifier	Sau reject/manual review	reject note, flags, OCR/face/liveness signals	NLP classification	reason category	Dễ evaluate, có ops analytics	Cần label synthetic
24	KYC reviewer decision consistency analysis	Review quality control	same case type, reviewer decision, risk signal	DS analytics	inconsistent decision report	Rất hay cho governance/human-in-loop	Cần mô phỏng reviewer behavior
25	KYC manual review workload prediction	Ops planning	daily case volume, risk mix, review duration	Forecasting/DS	workload forecast	Có DS thực tế, ít người nghĩ tới	Xa AI nếu không gắn review system

26	Drop-off prediction during onboarding	User onboarding funnel	step reached, retry count, document fail, device, time spent	ML classification	dropout risk + intervention	Gần product analytics, hữu ích	Không hoàn toàn KYC risk
27	Friction optimization for eKYC retries	Khi user bị fail KYC	fail reason, retry history, document quality	DS/ranking	best next instruction	Có product + AI angle	Cần metric rõ: completion rate giả lập
28	KYC instruction personalization	User cần retake/upload lại	fail reason, user language, device, error type	LLM templating + eval	personalized instruction	Hợp LLM nhưng scope nhỏ	Dễ quá nhẹ nếu đứng riêng
29	Business KYC / KYB merchant onboarding risk	Merchant onboarding	business registration, owner identity, merchant category, address, bank account	ML/rule risk scoring	merchant onboarding risk	Rất hợp ví/payment merchant	KYB domain khó hơn KYC cá nhân
30	Merchant document mismatch detection	KYB document review	business name, tax code, owner name, bank account name	NLP/rule comparison	mismatch flags	Cụ thể, hợp merchant onboarding	Data phải synthetic

31	KYC data quality dashboard	Data/ops monitoring	missing rate, OCR confidence distribution, rejection reason	DS analytics	data quality report	Rất tốt nếu mentor thích DS/ops	Không đủ ML nếu đứng riêng
32	KYC model drift monitoring	Sau khi deploy model giả lập	OCR confidence distribution, pass rate, rejection mix over time	DS drift detection	drift alert	Có AI governance mindset	Nên làm add-on, không làm chính
33	Bias/fairness check for KYC routing	Model review	review rate by region/device/document quality group	DS fairness analysis	fairness report	Tư duy responsible AI tốt	Nhạy cảm, cần synthetic groups
34	KYC decision audit trail design	Compliance logging	model version, rules hit, reviewer decision, timestamp	SE + DS audit analytics	audit trail	Hợp fintech governance	Không phải AI/ML core
35	KYC case similarity search	Reviewer cần xem case tương tự	embeddings of case signals/reasons, previous outcomes	Vector search / ML embeddings	similar past cases	Hợp AI search, hữu ích reviewer	Cần build dataset case history giả lập
36	KYC knowledge-base RAG assistant	Reviewer hỏi policy	KYC policy docs, case facts	RAG + citation + eval	grounded answer	Hợp AI agent/eval	Chỉ hay nếu có policy KB rõ

37	KYC policy violation checker	Kiểm tra quyết định reviewer	case facts, policy rules, decision	Rule + LLM judge/eval	possible policy violation	Có governance angle mạnh	LLM judge phải cẩn thận, dễ sai
38	Explainable KYC risk report	Tổng hợp case cho compliance	all risk signals, decision, evidence	LLM structured report + rules	compliance-friendly report	Gộp tốt với exception review	Nếu quá nhiều template thì ít ML
39	KYC fraud ring early signal	Onboarding graph monitoring	shared devices, IPs, addresses, documents	Graph anomaly	suspicious onboarding cluster	Có chiều sâu, khác biệt	Lại gần fraud, data khó
40	KYC synthetic data generator for testing	Test/evaluation setup	scenario templates, mismatch types, confidence distributions	Generative/simulation + DS	synthetic labeled KYC dataset	Rất hữu ích nếu thiếu data thật	Nếu chỉ generator thì ít product value

Package 1 — AI-Assisted eKYC Exception Review

Gồm:

- #1 eKYC exception review scoring
- #2 Form-vs-document mismatch detection
- #19 KYC risk banding
- #21 Reviewer checklist generator
- #22 Audit summary generation

Package 2 — KYC Identity Risk & Duplicate Detection

Gồm:

- #13 Duplicate identity detection
- #14 Multi-account device reuse detection
- #15 Synthetic identity risk scoring
- #35 Case similarity search
- #39 Fraud ring early signal

Package 3 — KYC Ops Quality & Governance

Gồm:

- #23 Rejection reason classifier
- #24 Reviewer decision consistency analysis
- #31 Data quality dashboard
- #32 Model drift monitoring
- #33 Bias/fairness check
- #34 Decision audit trail